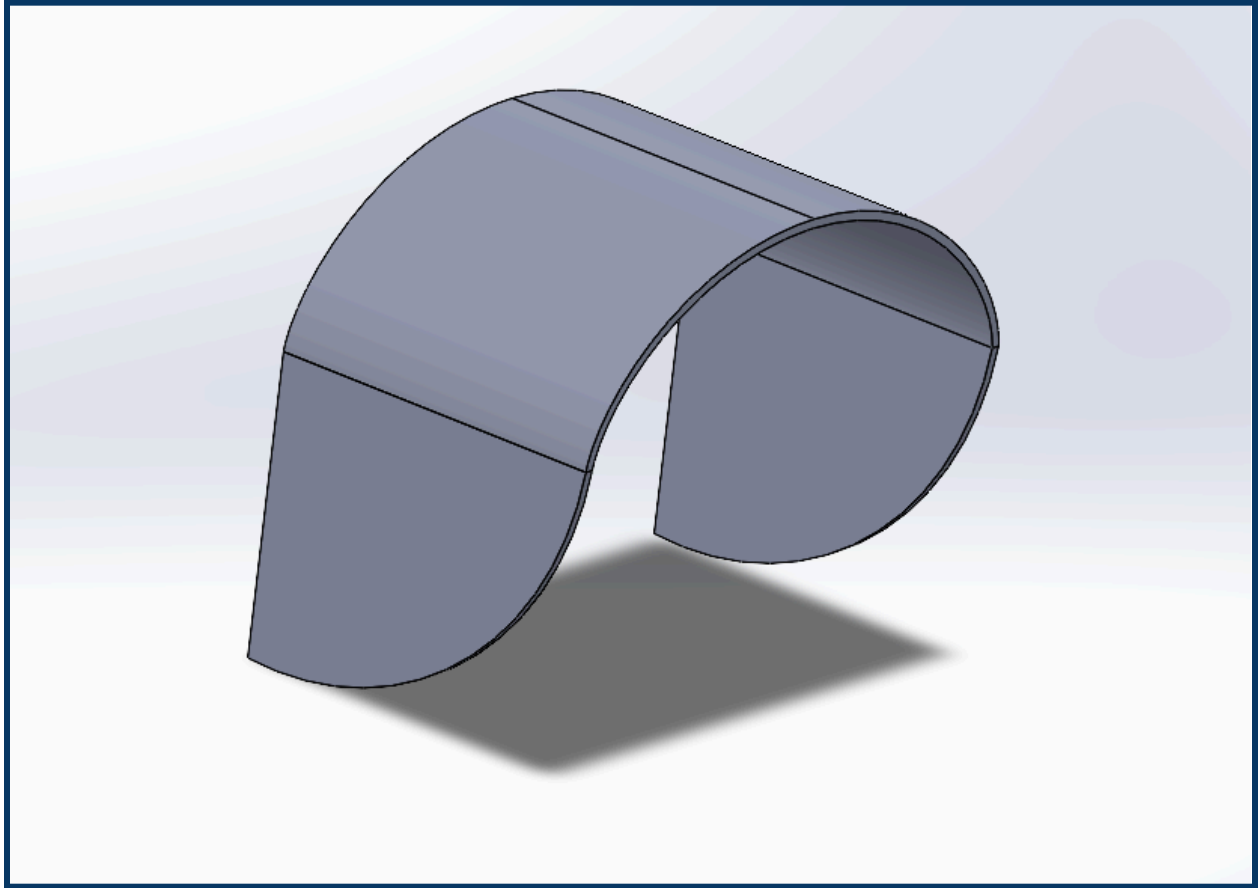


Re-engineering of Traffic light

MEE270 Lab report



Joseph Danilulk

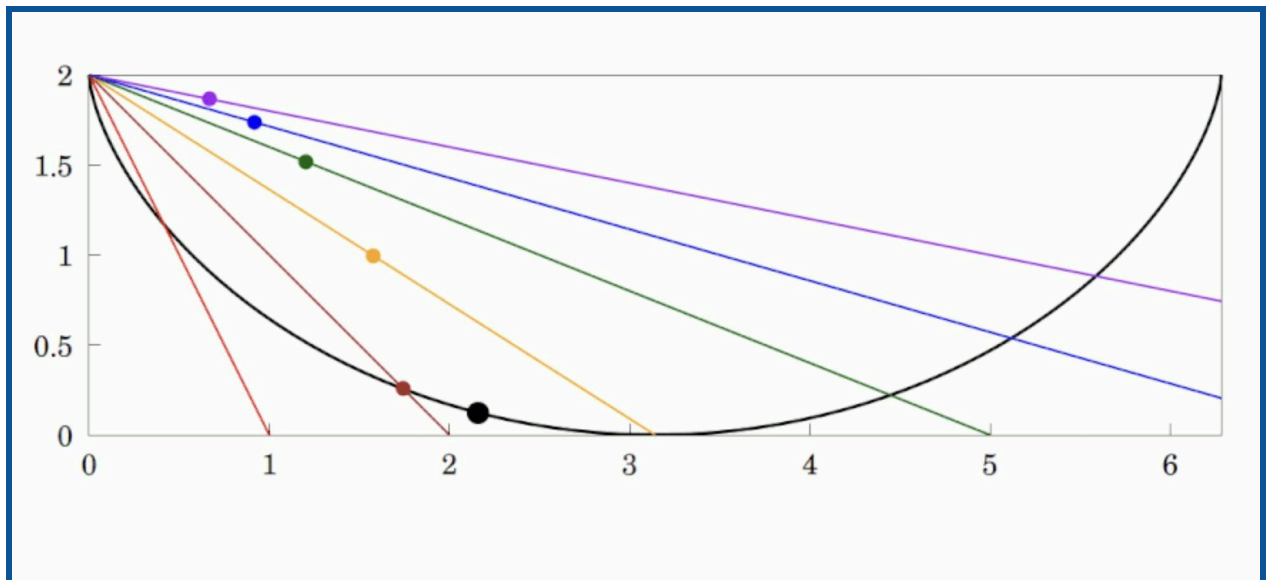
4/17/2025
(Final Report)

INTRODUCTION

This project explores the re-engineering of a traffic light visor to address the problem of snow accumulation in cold-weather regions. A passive, maintenance-free solution was developed using a flipped brachistochrone curve to help snow shed quickly from the visor surface, improving traffic signal visibility and safety.

HYPOTHESIS

By applying the flipped brachistochrone curve (the path of least time), I hypothesize that snow and ice will slide off the visor more quickly than on traditional designs, reducing buildup and improving visibility in adverse weather.

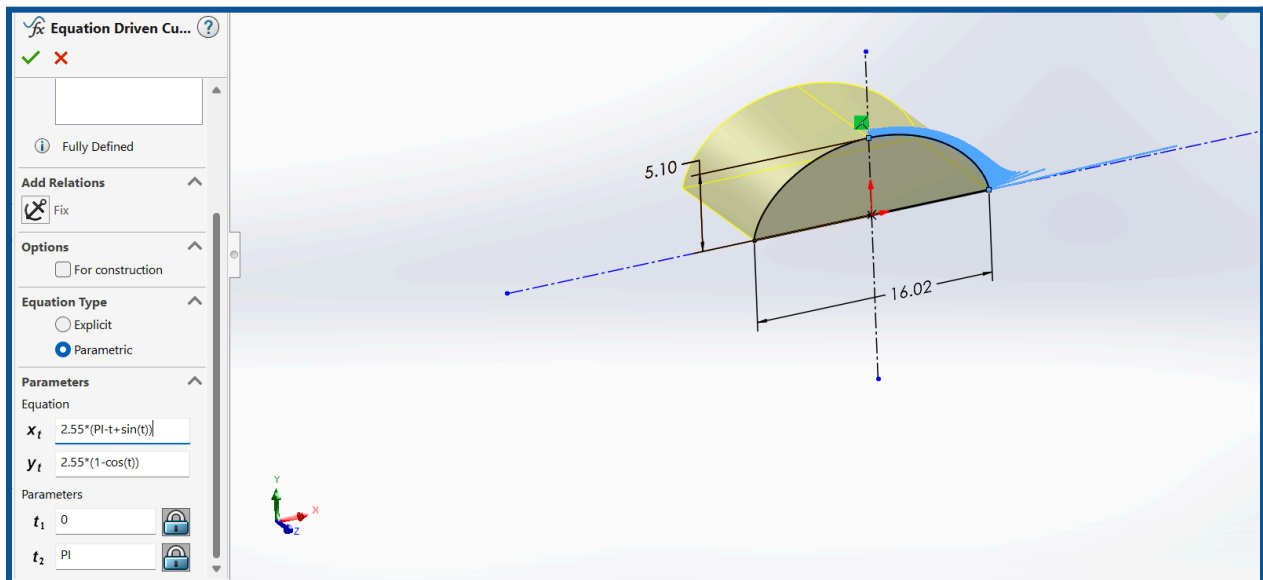


PROCEDURE

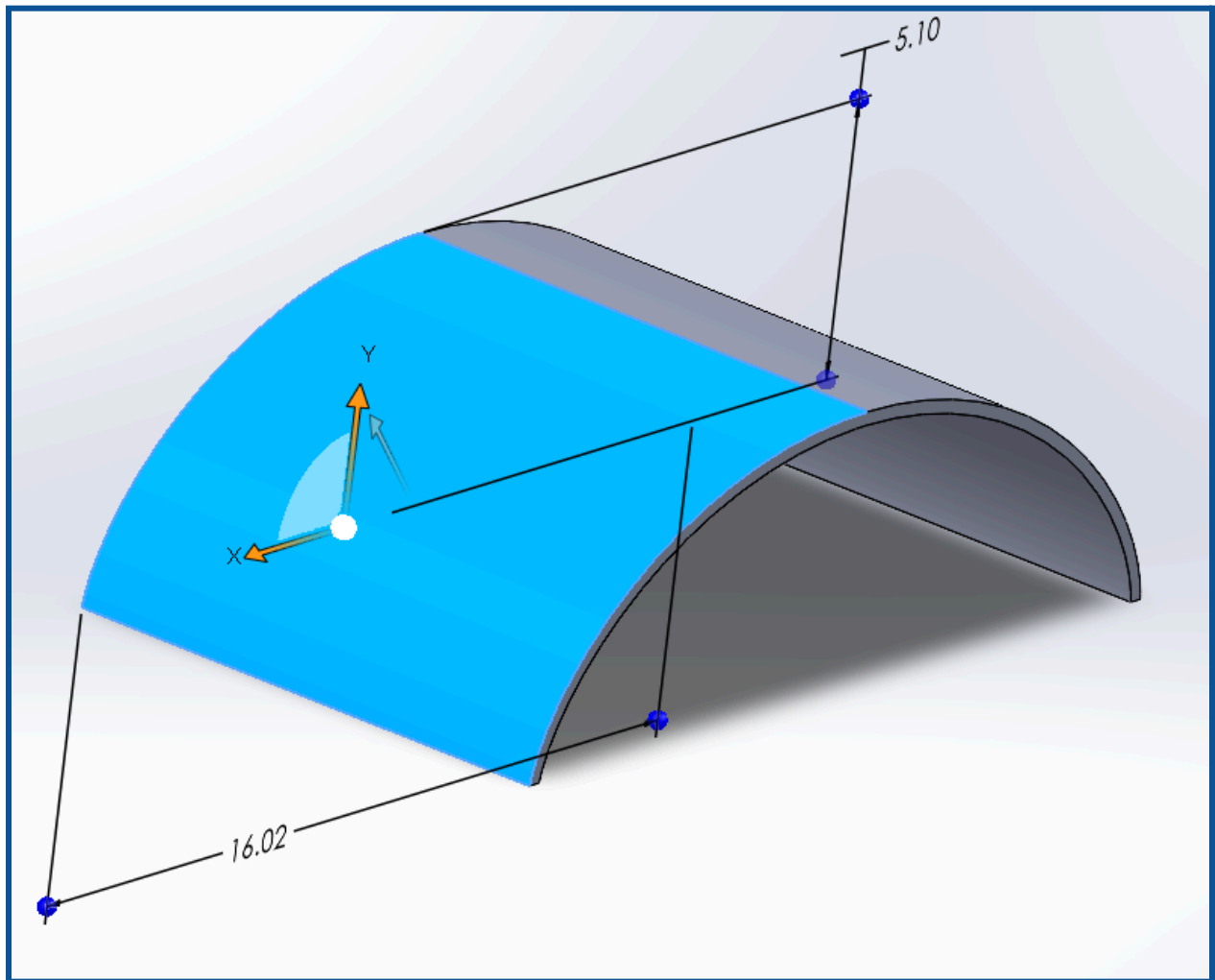
Using the **brachistochrone curve equation**:

- $x(t) = 2.55(\pi - t + \sin t)$
- $y(t) = 2.55(1 - \cos t)$, with $t \in [0, \pi]$

1. The curve is entered into SolidWorks using an **equation-driven spline**.
2. I Mirrored the curve about the vertical axis to form a symmetrical visor shape.

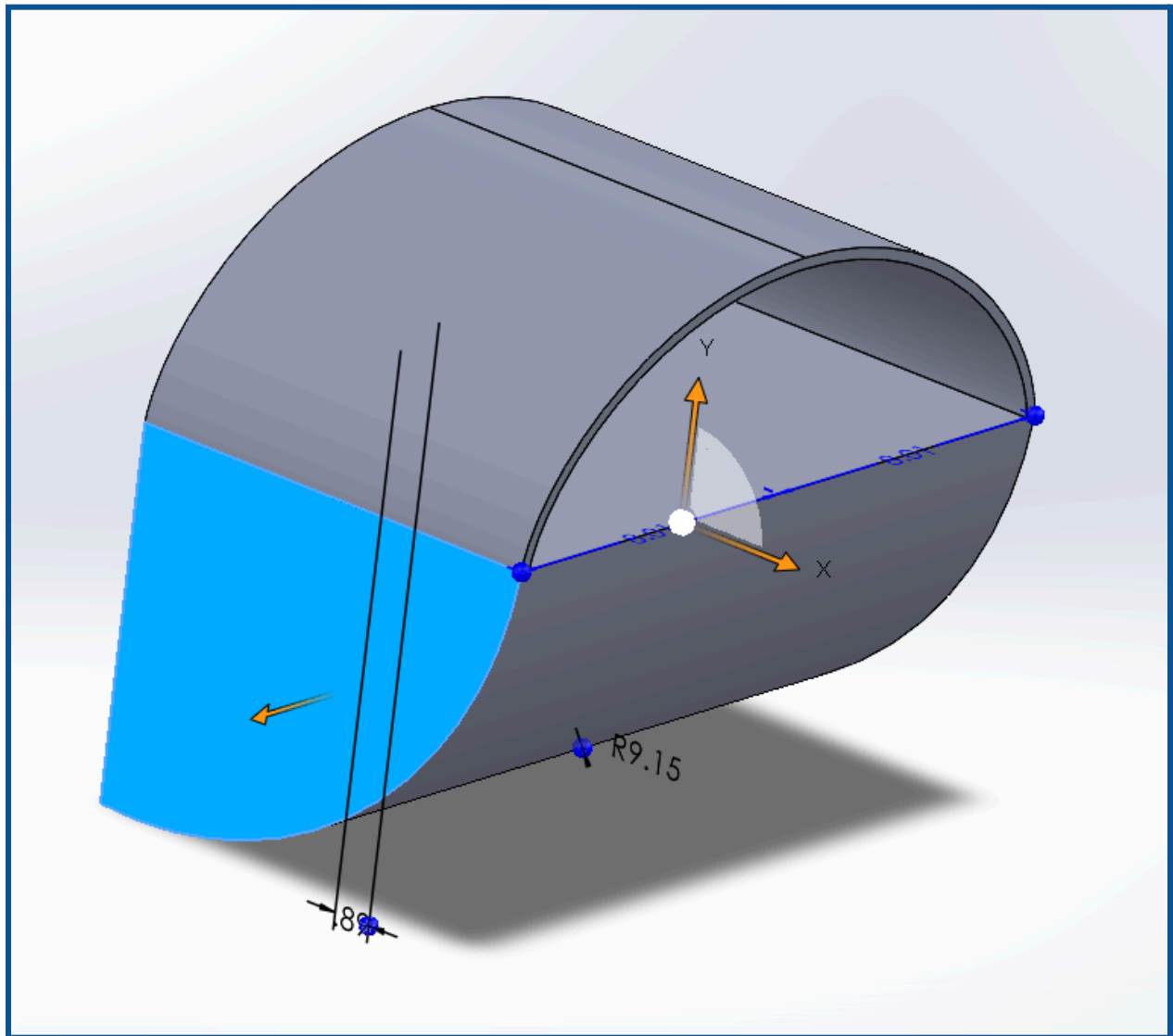


3. The curve is then extruded to create a 3D hood with a depth of **10 inches**.
4. **Offset Entities** (0.25 in) and the **Cut-Extrude** tool to hollow out the interior and define wall thickness.

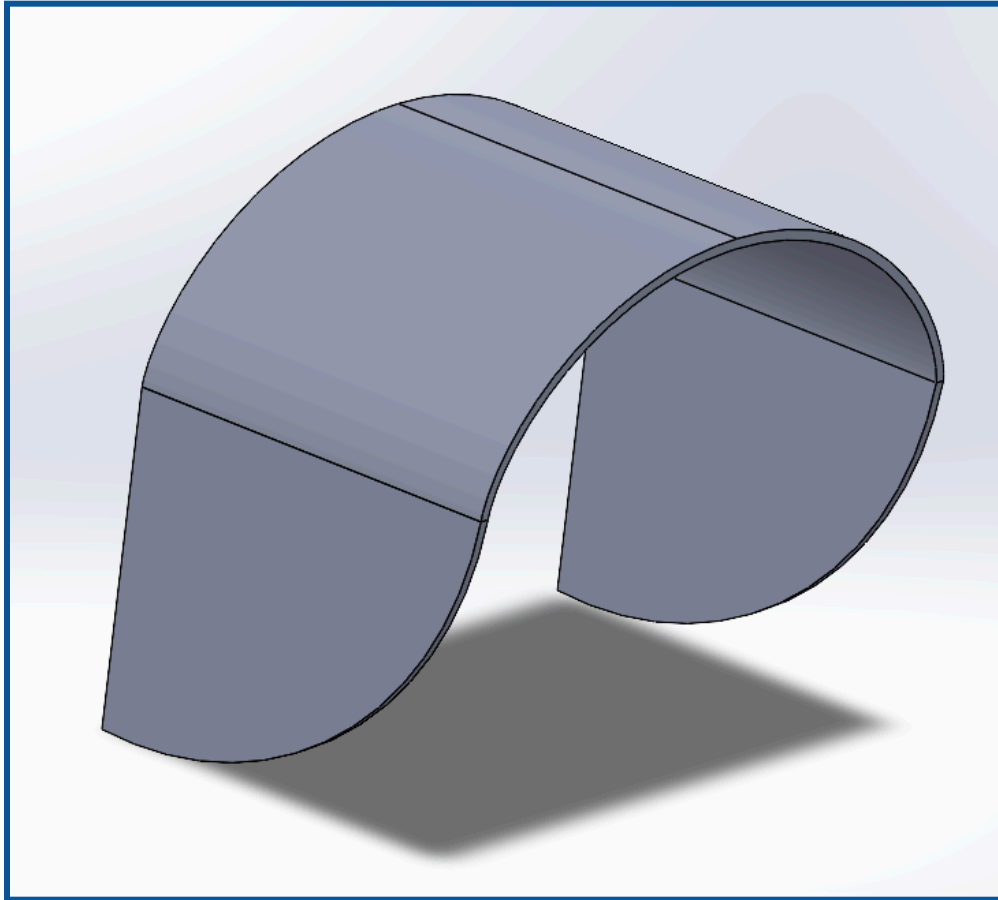


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5. Design of side panels was done by using half of the curve's profile and extruding them through the full depth.

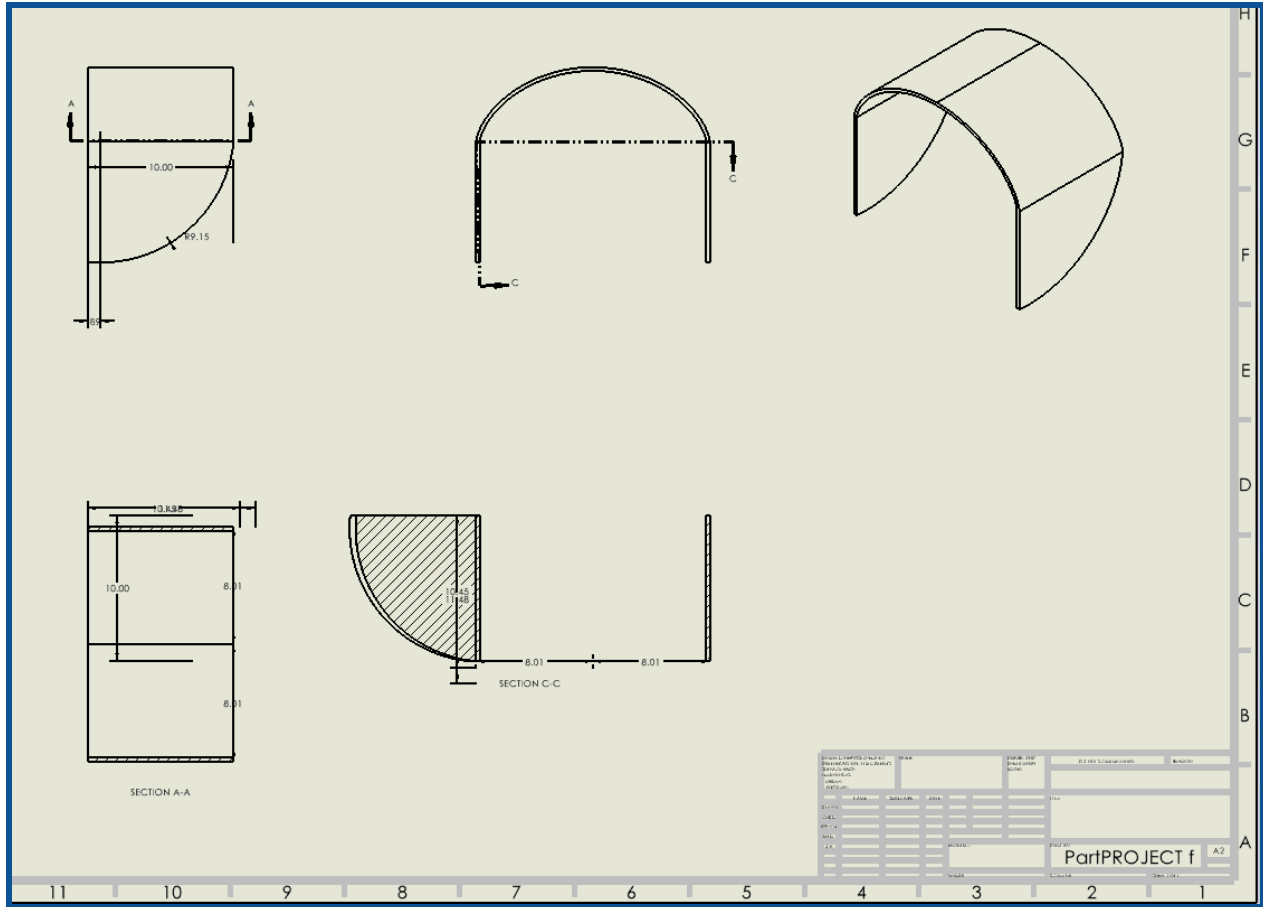


6. **Cut-Extrude** tool to hollow out the interior and define wall thickness.



Dimensions:

- Visor Width: 16 inches
- Depth: 10 inches
- Maximum Height: 5.10 inches above lens center
- Wall Thickness: 0.25 inches
- Mount clearance based on standard 12" signal housing
- **Curve Equation:**
- Parametric:
 $x(t)=2.55(\pi-t+\sin t)$
 $y(t)=2.55(1-\cos t)$



Technical Drawing: The 2D technical drawing of the visor communicates the geometry and key dimensions required for manufacturing. The drawing includes top, front, side, and isometric views, along with labeled dimensions for width (16"), depth (10"), height (~5.10"), and wall thickness (0.25"). The curve was defined using an equation-driven spline based on the flipped brachistochrone, and dimensions were added according to drafting standards.

CONCLUSION: I believe the design effectively addresses the challenge of snow accumulation on traffic lights using a passive approach. The design allows for minimal snow retention and promotes fast sliding off of snow and ice particles, to help improve traffic safety during winter conditions. The final model satisfies the functional goals and can be manufactured without the need for electrical components or complex maintenance.